

Rajendra Bist

Full-Stack Web Developer & Backend Engineer

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SUMMARY

Full-stack engineer with a backend-first mindset, trained at Vcare Technical Institute and currently pursuing B.Tech (Computer Science) at Indira Gandhi National Open University. Designs and ships AI-integrated, production-grade systems — from WebSocket event architecture to streaming LLM pipelines — with deliberate attention to API design, database schema modelling, and system reliability. Three deployed AI and real-time products in production. Seeking backend-heavy full-stack or AI engineering roles.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), TypeScript, Python (basic), HTML5, CSS3
Frontend:	React 18, Next.js 15 (App Router), Tailwind CSS, Framer Motion, Responsive Design
Backend:	Node.js, Express.js, REST API Design, WebSockets (Socket.io), JWT Auth, Middleware Architecture
Databases:	MongoDB Atlas, Mongoose ODM, Supabase (PostgreSQL), Database Schema Design, TTL Indexing
AI / LLM:	Gemini API, Groq API, Streaming Interfaces, Prompt Engineering, Clerk Auth, Vercel AI SDK
Tools:	Git, GitHub, Vercel, Railway, Docker (learning), Linux CLI, GitHub Actions (learning)

PROJECTS

EduMethod AI — AI-Powered Personalised Learning Platform

Next.js · TypeScript · Tailwind CSS · Clerk · Supabase · Groq API · Gemini API | [GitHub](#) | Live: edumethod-ai.vercel.app

- Architected a full-stack AI learning platform that converts any book photo or syllabus text into a structured 7-day study plan with spaced repetition scheduling, adaptive quizzes, and a conversational AI doubt-coach — end-to-end in a single user flow.
- Integrated Groq API (Llama) for low-latency study plan generation and Gemini API for multimodal syllabus extraction from uploaded book photos; engineered custom prompt chains to produce structured JSON learning paths consumed by the frontend.
- Production auth and data layer: Clerk for authentication (session management, protected routes), Supabase (PostgreSQL) for persistent user progress tracking and spaced repetition state — zero data loss on refresh.

SocraticAI — Socratic Method Learning Assistant

Next.js 15 · Node.js · Gemini API · MongoDB · Tailwind CSS | [GitHub](#) | Live: socratic-ai-tau.vercel.app

- Built and deployed a chat-first AI assistant that enforces Socratic dialogue via custom system prompt engineering — countering user questions with guided reasoning prompts rather than direct answers, producing measurably deeper engagement.
- Streaming architecture: Gemini API response streamed token-by-token via Web API ReadableStream in a Next.js 15 serverless route; MongoDB Atlas logs sessions with TTL auto-expiry; in-memory rate limiting protects free-tier quota.

WhatsApp Clone — Real-Time Messaging System

React · Node.js · Express.js · Socket.io · Tailwind CSS | Live: whats-app-clone-client-liart.vercel.app

- Designed a bidirectional WebSocket event protocol (join-room, send-message, typing-status, disconnect) with Socket.io room isolation — handling concurrent multi-user sessions without message loss or ordering violations.
- Decoupled frontend (Vercel) and backend (Railway) with CORS-aware configuration; reflects production multi-service deployment architecture.

Developer Portfolio with Streaming AI Chat Agent

Next.js 15 · Gemini API · MongoDB · Framer Motion | Live: rajendra-bist.vercel.app

- Floating AI agent streams context-aware portfolio responses (Gemini 1.5 Flash) via ReadableStream; dark/light theme system, Framer Motion scroll-reveal, contact form with MongoDB persistence — Lighthouse Performance ≥ 90.

EDUCATION

B.Tech in Computer Science & Engineering (Ongoing)

Indira Gandhi National Open University (IGNOU)

Full-Stack Web Development

Vcare Technical Institute · Next.js, Node.js, React, MongoDB, Express.js, REST APIs, PostgreSQL, System Design

+2 Science

KMC — Kailali Multiple Campus

Secondary Education (SEE)

Shree Phulwari Secondary School

ADDITIONAL

- Currently Building:** EduMethod AI — extending adaptive quiz engine and spaced-repetition analytics dashboard
- Backend Focus:** System design patterns, JWT auth flows, microservice architecture, PostgreSQL query optimisation
- Location:** Nepal · UTC+5:45 — available for remote worldwide; open to onsite/hybrid in Nepal or India
- Languages:** Nepali (Native), Hindi (Fluent), English (Professional)